

CH5440 ENDSEM REPORT
ME20B072

- 1)
- a) By applying PCA and finding the zero eigenvalues of the covariance matrix we can find the number of constraints, in our case the number of constraints is 9
This is because the rows of the constraint matrix lie in the null space of the data matrix
 - b) To get a valid set of dependent variables the Ad constructed from the columns of constraint matrix corresponding to the selected dependent variables, now this matrix should be invertible to get the solution for the dependent variables from the independent variables, so we find the matrix with the least possible condition number amongst all the combinations
 - c) By performing hypothesis testing we can see that the correct number of constraints are being identified, the table is displayed in the code
 - d) the maximum absolute difference between the regression matrix is 0.0198327
- 2) for this question the necessary explanations are provided in the code